Roe Notes

Contents

1 Riemannian Geometry

1.1 Connections

Let M be a smooth manifold, and let V be a vector bundle on M.

Definition 1.1. A *Connection* on V is a linear map

$$\nabla : C^\infty(TM) \otimes C^\infty(V) \to C^\infty(V)$$

which assigns to a vector field X and a section Y of V a new section $\nabla_X Y$ of V, such that for any smooth function f on M,

1. $\nabla_{fX} Y = f\nabla_X Y$
2. $\nabla_X fY = f\nabla_X Y + (X.f)Y$,

where $X.f$ denotes the lie derivative of f along X.

Condition (1.) tells us that $X \mapsto \nabla_X Y$ is a homomorphism of modules over $C^\infty(M)$. In other words, a connection is tensorial in $C^\infty(TM)$. This means that we can regard a connection as a map

$$C^\infty(V) \to \Omega^1(V) := C^\infty(T^*M \otimes V),$$

the space of V valued forms on M.

In a local trivialization, an example of a connection is the Lie derivative. Thus many local connections exist.

Moreover, the difference of two connections is local in both X and Y, making it an $\mathrm{End}(V)$ valued one-form. Thus, we can use partitions of unity to piece together local connections into a global one. So connections exist and are plentiful.

From the above discussion, we deduce that in a local trivialization,

$$\nabla_i = \frac{\partial}{\partial x_i} + \Gamma_i,$$

where $\nabla_i = \nabla_{\frac{\partial}{\partial x_i}}$, and Γ_i is a smooth section of $\mathrm{End}(V)$. The Γ_i are called the Christoffel symbols.

Remark 1.1. We can also define connections using *paralell transport.* Let γ be a smooth curve in M. The differential equation $\nabla_{\dot\gamma} Y = 0$ is a first order ODE on sections Y of V along γ which has a unique solution for a given initial value. One says that Y is paralell along γ, or that Y has been obtained from its initial value by paralell transport.

Definition 1.2. The *curvature operator* K of a connection ∇ on V is defined as follows: if X and Y are vector fields and Z is a section of V, then

$$K(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

Lemma 1.1. *The curvature $F(X,Y)Z$ at a point $m \in M$ depends only on the values of X, Y, and Z at the point m.*

Proof: Choose some local coordinate patch and a trivialization of V. Then we can expand

$$\begin{aligned} \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z = \sum_{i,j} &\left[X^j \frac{\partial Y^i}{\partial x_j} - Y^j \frac{\partial X^i}{\partial x_j} \right] \frac{\partial Z}{\partial x_i} + \left[X^j Y^i - Y^j X^i \right] \frac{\partial^2 Z}{\partial x_j \partial x_i} && (1) \\ &+ \left[X^j Y^i - Y^j X^i \right] \left(\frac{\partial \Gamma_j(Z)}{\partial x_i} + \frac{\partial \Gamma_i(Z)}{\partial x_j} \right) && (2) \\ &+ \left[X^j \frac{\partial Y^i}{\partial x_j} - Y^j \frac{\partial X^i}{\partial x_j} \right] (\Gamma_i(Z)) + \left[X^j Y^i - Y^j X^i \right] (\Gamma_j \Gamma_i(Z)). && (3) \end{aligned}$$

Since mixed partials commute, the second term in (1) vanishes. By symmetry, the term (2) vanishes as well, leaving only

$$\begin{aligned} \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z &= \sum_{i,j} \left[X^j \frac{\partial Y^i}{\partial x_j} - Y^j \frac{\partial X^i}{\partial x_j} \right] \left(\frac{\partial Z}{\partial x_i} + \Gamma_i(Z) \right) \\ &\quad + \left[X^j Y^i - Y^j X^i \right] (\Gamma_j \Gamma_i(Z)) \\ &= \nabla_{[X,Y]} Z + \left[X^j Y^i - Y^j X^i \right] \Gamma_j \Gamma_i(Z). \end{aligned}$$

From here the lemma follows immediately. □

Thus K is induced by a bundle map $TM \otimes TM \to \mathrm{End}(V)$, and moreover since $K(X,Y)$ is antisymmetric in X and Y, we may regard K as a 2-form with values in $\mathrm{End}(V)$. In terms of local coordinates, we define the curvature 2-form to be the $\mathrm{End}(V)$-valued 2-form

$$K = \sum_{i<j} K(\partial_i, \partial_j) dx^i \wedge dx^j,$$

where $K(\partial_i, \partial_j)$ takes values in $\mathrm{End}(V)$.

Now we consider connections on the tangent bundle TM. Let x^i be local coorinates with a corresponding vector fields ∂_i. Then we can express the endomorphisms Γ_i of remark (1.1) as matrices:

$$\nabla_i(\partial_j) = \sum_k \Gamma^k_{ij} \partial_k.$$

These symbols completely determine the connection.

Definition 1.3. A connection ∇ on TM is called *symmetric* if for any two vector fields X and Y on TM, we have

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

1.2 Riemannian Geometry

Suppose M is a Riemannian manifold.

Definition 1.4. We say ∇ is *compatible with the metric* on M if for any three vector fields X, Y_1, and Y_2, we have

$$(\nabla_X Y_1, Y_2) + (Y_1, \nabla_X Y_2) = X \cdot (Y_1, Y_2).$$

Theorem 1.1 (Levi-Civita). *A Riemannian manifold possesses a unique connection that is symmetric and compatible with the metric.*

Proof: Work in local coordinates. Represent the metric by a matrix of smooth functions $g_{jk} = (\partial_j, \partial_k)$. The symmetry condition implies

$$(\nabla_i \partial_j, \partial_k) + (\partial_j, \nabla_i \partial_k) = \partial_i (\partial_j, \partial_k),$$

or in other words,

$$\partial_i g_{jk} = \sum_a \left(\Gamma^a_{ij} g_{ak} + \Gamma^a_{ik} g_{aj} \right).$$

Moreover, we can permute indices to get

$$\partial_j g_{ki} = \sum_a \left(\Gamma^a_{jk} g_{ai} + \Gamma^a_{ji} g_{ak} \right) \qquad \partial_k g_{ij} = \sum_a \left(\Gamma^a_{ki} g_{aj} + \Gamma^a_{jk} g_{ai} \right).$$

Combining these and using symmetry, we see that

$$\sum_a \Gamma^a_{ij} g_{ak} = \frac{1}{2} \left(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij} \right).$$

This determines the Γ's uniquely, since g is invertible. □

Since the metric determines the Levi-Civita connection, it also uniquely determines the curvature.

Definition 1.5. The curvature of the Levi-Civita connection on a Riemannian manifold is called the *Riemann curvature operator* and is denoted by R. In components relative to a frame $\{e_i\}$ we write

$$R(e_j, e_k) e_l = \sum_i R^i_{ljk} e_i.$$

We also define the 4-covariant version of the Riemannian curvature tensor defined by

$$R_{iljk} = (R(e_j, e_k)e_l, e_i).$$

Proposition 1.1. *The Riemann curvature operator has the following properties:*

(i) $R(X,Y)Z + R(Y,X)Z = 0$, *or* $R^i_{ljk} + R^i_{lkj} = 0$.

(ii) $R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0$, *or* $R^i_{ljk} + R^i_{jkl} + R^i_{klj} = 0$.

(iii) $(R(X,Y)Z,W) + (R(X,Y)W,Z) = 0$, *or* $R_{iljk} + R_{lijk} = 0$.

(iv) $(R(X,Y)Z,W) = (R(Z,W)X,Y)$, *or* $R_{iljk} = R_{jkil}$.

The only non-trivial contraction of the Riemann curvature tensor is called the *Ricci curvature tensor*, given by

$$\mathrm{Ric}(Y,Z) \mapsto \mathrm{tr}\,(X \mapsto R(X,Y)Z)\,.$$

In components, $\mathrm{Ric}_{ab} = \sum_i R^i_{aib}$.

We can raise one index of the Ricci curvature tensor to get the Ricci curvature operator, and taking its trace we obtain the scalar curvature,

$$\kappa = g^{ab}\mathrm{Ric}_{ab}.$$

Definition 1.6. We say a curve γ is a *geodesic* if $\nabla_{\dot\gamma}\dot\gamma = 0$, or in other words if $\dot\gamma$ is parallel along γ.

The geodesic equation has a unique solution given initial values $\gamma(0)$ and $\dot\gamma(0)$. Then the exponential map

$$\exp : U \to M$$

can be defined at any $m \in M$ on some open star-shaped subset $U \subset T_mM$ by sending a vector $v \in T_mM$ to the value at $t = 1$ of the unique geodesic with starting value m and velocity v. By the inverse function theorem, exp is a diffeomorphism from a neighborhood of zero in T_mM to a neighborhood of m in M. Choosing an orthonormal basis for T_mM gives a coordinate system called a *geodesic coordinate system* at m.

Proposition 1.2. *At the origin of a geodesic coordinate system, the Christoffel symbols all vanish.*

Proof: It is enough to show that $\nabla_i\partial_j$ vanishes at the origin. By symmetry of the connection, it is also enough to show $\nabla_X X = 0$ for all constant vector fields X.

But in a geodesic coordinate system, the geodesics are radial lines from the origin. Moreover, X clearly points along these geodesics and has constant length, so $\nabla_X X = 0$. □

1.3 Differential forms

We let $\Omega^p = \bigwedge^p V^*$ denote the space of p-forms. We can identify p-forms with antisymmetric tensors by writing

$$\alpha = \sum_{i_1<\cdots<i_p} A_{i_1,\ldots,i_p} dx^{i_1} \wedge \cdots \wedge dx^{i_p} = \frac{1}{p!} \sum_{i_1,\ldots,i_p} A_{i_1,\ldots,i_p} dx^{i_1} \wedge \cdots \wedge dx^{i_p}.$$

We first extend the metric to the space of contravariant tensors.

Let $(\cdot,\cdot)$ be an inner product on V, and fix a basis $\{e^i\}$ of V, with corresponding dual basis $\{e_i\}$. We have the matrix $g_{ij} = (e_i, e_j)$ and its inverse g^{ij}.

Then we have the lowering and raising of incices maps,

$$\begin{aligned} \flat &: e^i \mapsto g_{ij} e_j \\ \sharp &: e_i \mapsto g^{ij} e^j. \end{aligned}$$

These extend to tensors in the natural way.

Using the metric, we can define one on the space of $(0,p)$ tensors by the formula

$$(\alpha,\beta) = \frac{1}{k!} \sum_{i_1,\ldots,i_p,j_1,\ldots,j_p} g^{i_1,j_1} \cdots g^{i_p,j_p} A_{i_1,\ldots,i_p} B_{i_1,\ldots,i_p}.$$

In an oriented coordinate system, we define the volume form $\mathrm{vol} = \sqrt{g} dx^1 \wedge \cdots dx^n$, where $g = \det g_{ij}$. The normalization constant in the inner product formula ensures that $(\mathrm{vol},\mathrm{vol}) = 1$:

$$\begin{aligned} (\mathrm{vol},\mathrm{vol}) &= \frac{1}{n!} \sum_{i_1,\ldots,i_n,j_1,\ldots,j_n} g^{i_1,j_1} \cdots g^{i_n,j_n} \delta^{i_1,\ldots,i_n}_{1,\ldots,n} \delta^{i_1,\ldots,i_n}_{1,\ldots,n} \\ &= \frac{1}{n!} g \sum_{j_1,\ldots,j_p} \det(g^{ij}) = 1. \end{aligned}$$

We also have formulas for the wedge product and exterior derivative of forms in terms of the antisymmetric tensors associated to them. If C corresponds to $\alpha \wedge \beta$, then

$$C_{k_1,\ldots,k_{p+q}} = \frac{1}{p!q!} \sum_{i_1,\ldots,i_p,j_1,\ldots,j_q} \delta^{i_1,\ldots,i_p,j_1,\ldots,j_q}_{k_1,\ldots,k_{p+q}}.$$

and if $D \sim d\alpha$, then

$$D_{k_1,\ldots,k_{p+1}} = \frac{1}{p!} \sum_{i_1,\ldots,i_p} \delta^{j,i_1,\ldots,i_p}_{k_1,\ldots,k_p} \frac{\partial A_{i_1,\ldots,i_p}}{\partial x_j}.$$

Definition 1.7. Let α be a k-form. Define the *Hodge Star* $*\alpha$ to be the unique $(n-k)$ form such that for all k-forms β,

$$(\alpha,\beta)\mathrm{vol} = \beta \wedge *\alpha.$$

This operation is linear and we have $**\alpha = (-1)^{kn+k}\alpha$

Definition 1.8. If α is a k-form, define

$$d^*\alpha = (-1)^{nk+n+1} * d * \alpha.$$

Then $d^*\alpha$ is a $(k-1)$ form, and $(d^*)^2 = 0$.

The importance of d^* is that it is the adjoint of d, once we define a global inner product on forms.

We define

$$\langle \alpha, \beta \rangle = \int_M (\alpha, \beta)\mathrm{vol} = \int_M \beta \wedge *\alpha = \int_M$$

1.4 Excercises

1.28 Prove Cartan's formula for the exterior deriative of a p-form.

Proof: See homework either 6 or 7 from DG.

1.29 Prove the second Bianchi Identity,

$$\nabla R(X, Y, Z, V, W) + \nabla R(X, Y, V, W, Z) + \nabla R(X, Y, W, Z, V) = 0.$$

Proof: Since the objects are tensors, it suffices to prove the identity at an arbitrary point on M. Fix a geodesic coordinate system about this point so that brackets and $\nabla_i e_j$ vanishes at the origin.

Then follow the steps in Do Carmo with the simplified coordinate system, then the result is clear.

1.30 Let x^i be a geodesic coordinate system on a manifold M, and let r denote the Riemannian distance from the point x to the origin. Prove that

$$\sum_{i,j} g_{ij}(x)x^i x^j = \sum_{i,j} g^{ij}(x)x^i x^j = r^2.$$

Proof: This problem is straightforward using the fact that in a geodesic coordinate system, the geodesics are just radial lines from the origin. The distance from the origin to x is the infimum over all geodesics connecting these points of the length of these curves. But in this case, there is only one radial line, so its length is the inf.

Moreover, the velocity of the geodesic at the origin at $t = 0$ and at x at $t = 1$ is just the magnitude of the vector x, viewed as a subset of T_0M.

1.31 Let α be a k-form on an n -dimensional oriented Riemannian manifold M represented in local coordinates by an antisymetric tensor A. Show that

$$*\alpha = \sum_{i_1,\ldots,i_k,j_1,\ldots,j_{n-k}} \frac{1}{(n-k)!k!} \delta^{1,\ldots,n}_{i_1,\ldots,i_k,j_1,\ldots,j_{n-k}} \sqrt{g} A^{i_1,\ldots,i_k} dx^{j_1} \wedge \cdots \wedge dx^{j_{n-k}}.$$

Proof: We utilize the antisymmetric tensor representation formula for the wedge product:

$$C_{k_1,\ldots,k_{p+q}} = \frac{1}{p!q!} \sum_{i_1,\ldots,i_p,j_1,\ldots,j_q} \delta^{i_1,\ldots,i_p,j_1,\ldots,j_q}_{k_1,\ldots,k_{p+q}},$$

where C is the tensor representing the wedge of a p-form and a q-form.

We also have a formula for the inner product of k forms:

$$(\alpha,\beta) = \frac{1}{k!} \sum_{i_1,\ldots,i_p,j_1,\ldots,j_p} g^{i_1,j_1}\cdots g^{i_p,j_p} A_{i_1,\ldots,i_p} B_{i_1,\ldots,i_p}.$$

By definition of $*\alpha$, for all k-forms β, we have

$$(\alpha,\beta)\mathrm{vol} = \beta \wedge *\alpha.$$

In other words, if we write $*A$ and B for the antisymmetric tensors corresponding to $*\alpha$ and β, this means

$$\sqrt{g}\frac{1}{k!} \sum_{|I|=|J|=k} g^{I,J} A_I B_J = \frac{1}{(n-k)!k!} \sum_{|I|=k,|J|=n-k} \delta^{IJ}_{1,\ldots,n} B_I (*A)_J.$$

We start from the left. Notice this is just the raising of the indices of A:

$$A^J = \sum_{|I|=|J|=k} g^{I,J} A_I.$$

Then we have

$$(\alpha,\beta)\mathrm{vol} = \sqrt{g}\frac{1}{k!} \sum_{|J|=k} A^J B_J.$$

Now we need a trick with the generalized Kronoker-delta symbols. For k-tuples I, J and an $n-k$-tuple K, we have

$$\delta^{IK}_{1,\ldots,n}\delta^{JK}_{1,\ldots,n}\delta^I_J = 1.$$

To see why, consider either when I and J are opposite sign, or the same.

Then we can write

$$\begin{aligned}
\sqrt{g}\frac{1}{k} \sum_{|J|=k} A^J B_J &= \frac{1}{(n-k)!k!}\sqrt{g}\frac{1}{k!} \sum_{|I|=|J|=k,|K|=n-k} \delta^{IK}_{1,\ldots,n}\delta^{JK}_{1,\ldots,n}\delta^I_J B_J A^J \\
&= \frac{1}{(n-k)!k!}\sqrt{g}\frac{1}{k!} \sum_{|I|=|J|=k,|K|=n-k} \delta^{IK}_{1,\ldots,n}\delta^{JK}_{1,\ldots,n} B_J A^I \\
&= \frac{1}{(n-k)!k!} \sum_{|J|=k,|K|=n-k} \delta^{IJ}_{1,\ldots,n} B_I \left(\sqrt{g}\frac{1}{k!}\sum_{|I|=k} \delta^{IK}_{1,\ldots,n} A^I\right) \\
&= B \wedge *A,
\end{aligned}$$

where we write $B\wedge *A$ to be the tensor for $\beta \wedge *\alpha$. In other words, the matrix in parenthesis with parameter K is $*A$.

2 Connections, Curvature, and Characteristic Classes

Let G be a Lie group.

Definition 2.1. We define a *principal bundle E* with *structural group G* over a smooth manifold M to be a locally trivial fiber bundle whose fibers are homeomorphic to G, considered as a right G-space. Therefore, G acts smoothly on E by right multiplication on each fiber, and $M = E/G$.

Examples of these are frame bundles of vector bundles. These are principal $\mathrm{GL}(k)$-bundles.

On the other hand, given a principal bundle, there is a way to construct a vector bundle with a G action, called the *associated vector bundle* to E. Specifically, given a representation $\rho : G \to \mathrm{GL}(F)$ of G on a vector space F, G operates on the space $E \times F$ by

$$(e, f)g = (eg, \rho(g^{-1})f).$$

We define the associated vector bundle as the quotient space $E \times_\rho F$ under the G action. It is indeed a vector bundle whose fibers are isomorphic to F.

Now we will make some definitions about differential forms on vector bundles. Let E be a principal bundle with group G. To any $u \in \mathfrak{g}$, the Lie algebra of G, we can associate a G-invariant vector field on E. This is called the Killing Field.

Definition 2.2. Let $p \in M$ and choose $s \in E_p$. Consider the map $m_s : G \to E_p$ given by $g \mapsto sg$. This is in fact a diffeomorphism, so that for every $g \in G$, the differential $dm_s|_g : T_gG \to T_{gs}E_s$ is an isomorphism. Taking $g = e \in G$, we obtain an isomorphism $\varphi_p : T_eG = \mathfrak{g} \to T_sE_p$. For $u \in \mathfrak{g}$, we define the *Killing field X_u* to be the vector field on E whose value at a point $s \in E_p$ is φ_p

3 The Spin Groups

Definition 3.1. An algebra A over $\mathbb{R}$ or $\mathbb{C}$ is called a *superalgebra* if it is an internal direct sum of linear subspaces A_0 and A_1, with

$$A_0 \cdot A_0 \subset A_0, \quad A_0 \cdot A_1 \subset A_1, \quad A_1 \cdot A_0 \subset A_1, \quad A_1 \cdot A_1 \subset A_0.$$

We say that A_0 and A_1 are the *even* and *odd* parts of A respectively. The subset $A_0 \cup A_1$ is called the set of homogeneous elements of A.

Alternatively, a superalgebra is an algebra A equipped with an automorphism ε, the *grading automorphism*, such that $\varepsilon^2 = 1$. The subspaces A_0 and A_1 are the 1 and -1 eigenspaces of ε, and in the previous case the automorphism is defined by

$$\varepsilon(a_0 + a_1) = a_0 - a_1$$

for $a_0 \in A_0, a_1 \in A_1$.

Proposition 3.1. *The Clifford algebra of an inner product space V is a super algebra, in which the homogeneous elements are products of an even or odd number of elements of V.*

Definition 3.2. We define the *supercommutator* on A first on homogeneous elements by

$$[x, y]_s = xy - (-1)^{\deg(x)\deg(y)} yx,$$

and then extend to all of A by linearity. We define the *super-center* $\mathfrak{Z}_s(A)$ of A to be the set of all elements which *supercommute* with every element in A, i.e. $[x, y]_s = 0$.

4 Analytic Properties of Dirac Operators

This section covers Sobolev spaces, continuity results for (generalized) Dirac operators, and the functional calculus.

4.1 Sobolev spaces

We work on the n-torus $\mathbb{T}^n$. We will define the sobolev spaces $H^k(\mathbb{T}^n)$ for k any positive integer using a version of the Fourier transform. Let $f : \mathbb{T}^n \to \mathbb{C}$ be an integrable function. Define its Fourier transform $\widehat{f} : \mathbb{Z}^n \to \mathbb{C}$ as

$$\widehat{f}(\nu) = \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} f(x) e^{i\langle x,\nu\rangle} dx.$$

We define the Fourier Series of f to be the "formal" sum

$$\sum_{\nu \in \mathbb{Z}^n} f(\nu) e^{i\langle x,\nu\rangle}.$$

Here are some basic results:

Theorem 4.1 (Parseval's Theorem). *The Fourier Transform $\mathcal{F}$ is an isometry $\mathcal{F} : L^2(\mathbb{T}^n) \to L^2(\mathbb{Z}^n)$, where the second space is equipped with the (normalized) counting measure. In other words,*

$$\int_{\mathbb{T}^n} |f|^2 = (2\pi)^n \sum_{\nu} |\widehat{f}(\nu)|^2.$$

Theorem 4.2 (Fourier Inversion Theorem for L^2). *For $f \in L^2(\mathbb{T}^n)$, the Fourier series of f converges in the L^2 norm.*

This gives us a way to translate from functions on $\mathbb{Z}^n$ back to functions on $\mathbb{T}^n$.

Theorem 4.3 (Fourier Inversion Theorem for C^∞). *For $f \in C^\infty(\mathbb{T}^n)$, the Fourier series of f converges in the Frechet C^∞ topology to f. The Fourier coefficients $\widehat{f}(\nu)$ are rapidly decreasing.*

Definition 4.1. Let k be a positive integer. Define the *Sobolev k-inner product* on $C^\infty(\mathbb{T}^n)$ by the formula

$$\langle f_1, f_2 \rangle_k = (2\pi)^n \sum_{\nu} \widehat{f_1}(\nu) \overline{\widehat{f_2}(\nu)} (1 + |\nu|^2)^k.$$

This makes sense since both $\widehat{f_1}$ and $\widehat{f_2}$ are rapidly decreasing. The norm induced by this inner product is called the *Sobolev k-norm.*

We define the *k-th Sobolev Space* $H^k(\mathbb{T}^n)$ to be the completion of $C^\infty(\mathbb{T}^n)$ in the Sobolev k-norm.

Proposition 4.1. *There is a continuous inclusion $C^\infty(\mathbb{T}^n) \to W^k(\mathbb{T}^n)$ for all k.*

Theorem 4.4 (Sobolev Embedding Theorem). *For any integer $p > n/2$, there is a continuous embedding $W^{k+p} \to C^k$.*

Theorem 4.5 (Rellich's Theorem). *If $k_1 < k_2$, there is a compact inclusion $W^{k_2} \to W^{k_1}$.*

We define Sobolev spaces on manifolds using partitions of unity. While the Sobolev k-norm will depend on the chosen atlas, one can show the induced topology is independent of this choice.

4.2 Analysis of the Dirac Operator

Recall the Weitzenbock formula,

$$D^2 = \nabla^*\nabla + \mathcal{K},$$

where $\mathcal{K}$ is a certain curvature operator. In fact, we can define *generalized* Dirac operators to be operators D such that

$$D^2 = \nabla^*\nabla + B,$$

where B is any first order operator on S.

From the results of the previous section, we immediately have the estimate

$$||Ds||_0 \leq C||s||_1,$$

since D is a first order operator. For generalized Dirac operators, we have an "approximate converse":

Theorem 4.6 (Garding's Inequality). *Let D be a generalized Dirac operator on a compact manifold. Then there is a constant C such that for any $s \in C^\infty(S)$,*

$$||s||_1 \leq C(||s||_0 + ||Ds||_0).$$

More Generally, we have

Proposition 4.2. *For any integer $k > 0$ there is a constant C_k such that for any $s \in C^\infty(S)$,*

$$||s||_{k+1} \leq C_k(||s||_k + ||Ds||_k).$$

To prove both of these results, it suffices to consider s supported within some coordinate patch.

Now we need some unbounded operator theory. We have

Proposition 4.3. *The closure $\overline{G}$ of the graph G of the Dirac operator is also the graph of an operator, which we will denote by $\overline{D}$.*

The proof of this fact relies on the existence of a formal adjoint of D. The domain of $\widehat{D}$ consists of all $x \in L^2(S)$ such that there is a sequence x_j converging to x, and with Dx_j converging in $L^2(S)$ as well. By applying Garding's inequality, we get

$$||x_i - x_j||_1 \leq C(||x_i - x_j||_0 + ||Dx_i - Dx_j||_0) \to 0,$$

and thus the sequence x_j is cauchy in $||\cdot||_1$ and converges to some x'. Then since the L^2 topology is weaker than the H^1 topology, this implies $x_j \to x'$ in L^2, and thus $x' = x$ by uniqueness of limits. Thus $x \in H^1$.

Conversely, if $x \in H^1$, take some smooth x_j converging to x in H^1. Then since D is continuous $H^1 \to H^0$,

$$||Dx_i - Dx_j||_0 \leq ||x_i - x_j||_1 \to 0,$$

so that Dx_j converges in L^2. And x_j clearly converges to x in L^2 as well, so $x \in \mathrm{Dom}(\overline{D})$.

Proposition 4.4. *Suppose $x, y \in L^2(S)$ with $Dx = y$ weakly, i.e.*

$$\langle x, s\rangle = \langle y, D^* s\rangle \quad \textit{for all } s \in C^\infty(S).$$

Then $x \in H^1(S) = dom(\overline{D})$, and $\overline{D}x = y$.

Proof: To prove this, the book uses a Friedrichs' mollifier. We obtain some bounded sequence $x_\varepsilon \to x$ weakly as $\varepsilon \to 0$, and prove that also Dx_ε is bounded. Applying Garding's inequality, we see that x_ε is bounded in the H^1 norm, and thus converges in L^2 to x by Rellich's theorem 4.5. This proves $x \in H^1$.

Then $\overline{D}x = z$ for some z. To prove that $z = x$, it is enough show that z is a weak solution to D as well and then apply uniqueness of weak derivatives. □

Proposition 4.5. *The Kernel of D consists of smooth sections.*

Proof: We will prove inductively that $s \in H^k$ for all k, and the result will follow by the Sobolev embedding theorem. We have

$$||s_\varepsilon||_k \le C_k(||s_\varepsilon||_{k-1} + ||Ds_\varepsilon||_{k-1}) = C_k(||s_\varepsilon||_{k-1} + ||[D, F_\varepsilon]s||_{k-1}),$$

since $Ds = 0$. But F_ε and $[D, F_\varepsilon]$ form bounded families of operators. Then s_ε is a bounded sequence in H^k which converges in L^2 to s, and must converge weakly in H^k. Then $s \in H^k$ as desired.

4.3 The Functional Calculus

We will not prove the next theorem, but it is some version of a spectral theorem for unbounded self adjoint operators.

Theorem 4.7. *There is a direct sum decomposition of H into a sum of countably many finite dimensional eigenspaces H_λ, each consisting of smooth sections with eigenvalue λ. These eigenvalues form a discrete subset of $\mathbb{R}$.*

Definition 4.2. A bounded operator A on $L^2(S)$ is called a *smoothing operator* if there is a smooth *kernel* $k(p, q)$ on $M \times M$ with values in $Hom(S_q, S_p)$, such that

$$As(p) = \int_M k(p, q)s(q)dq.$$